Wireless Intrusion Detection and Monitoring System (WIDMS)

Wireless Network Security (CBS 2343)

Introduction

The major form of current communication is wireless networks (IEEE 802.11), yet because their signals are broadcast freely over the air, they are inherently unsafe. Unencrypted management frames are a serious vulnerability that may be readily exploited by attackers using tools like Python's Scapy to track devices or execute Deauthentication DoS attacks.

The Wireless Intrusion Detection and Monitoring System (WIDMS) was created as a real-time detection tool to mitigate these hazards. Built using Kali Linux, Python, and Wireshark, WIDMS enables users to visualize anomalies, log suspicious events, and receive alerts before significant damage occurs. The project functions as an instructional platform that connects theoretical security principles with proactive, practical implementation by fusing attack simulation with defensive monitoring.



The Purpose of the Project

The system focuses on three specific threat areas:

- Deauthentication (Deauth) DoS attacks: Detecting forged frames used to forcibly disconnect legitimate users.
- Excessive probe requests: Identifying patterns used for unauthorized device tracking.
- Hidden SSID activities: Monitoring and identifying interactions with hidden wireless networks.

By combining attack simulation with defensive monitoring, the project demonstrates how easily wireless networks can be compromised using readily available tools. It serves as an educational platform to bridge the gap between theoretical security and hands-on application, helping students and network administrators recognize vulnerabilities and potential breaches before they lead to significant damage. Ultimately, WIDMS aims to raise awareness and promote proactive monitoring in home and small office environments

Background: Wireless Vulnerabilities

- Vulnerability: IEEE 802.11 networks are intrinsically insecure due to their broadcast nature and the use of unencrypted management frames (Deauth, Probe Requests).
- The Threat: Attackers use tools like Python’s Scapy to spoof these frames, launching Deauthentication DoS attacks or tracking devices.
- Security Gap: While WPA2/WPA3 secures data, management frames remain unprotected unless 802.11w is enabled.
- WIDMS Solution: This project provides a practical, low-cost educational platform to monitor these real-time threats using open-source tools.

Scope and Focus

Objective: Real-time detection and monitoring of wireless attacks within a controlled laboratory environment

Key Focus Areas:

- Deauthentication DoS: Simulating and detecting forged frames that disconnect clients.
- Probe-based Tracking: Monitoring excessive requests used for device tracking.
- Hidden SSID Detection: Identifying activities related to hidden networks.

- Technical Setup: Two Kali Linux machines using Python (Scapy) for attack simulation and Wireshark in monitor mode for analysis.
- Limitations: Strictly limited to detection and monitoring; does not include prevention (802.11w) or production-ready implementation

Methodology and Technical Stack

- Environment: Isolated, sandboxed lab network to ensure safety and ethical compliance.
- Kali Linux: Central auditing platform hosting both attack simulation and monitoring modules.
- Python (Scapy): Custom scripts for low-level packet manipulation, enabling the crafting and injection of raw 802.11 frames.
- Wireshark: Primary Deep Packet Inspection (DPI) tool used in monitor mode to capture and decode raw RadioTap headers.
- Telemetry Interface: Lightweight CLI with asynchronous logging for real-time alerts, attack statistics, and security event tracking.

Significance of the Project

- Addresses Critical Vulnerabilities: Focuses on the exploitation of unprotected management frames, one of the most common wireless security gaps.
- Essential Protection: In an era of universal wireless connectivity, proactive monitoring is vital to ensure data protection and network availability against Deauth DoS and tracking.
- Accessible Learning: Offers a low-cost educational solution using open-source tools (Kali Linux, Python Scapy, and Wireshark).
- Theory to Practice: Effectively bridges the gap between theoretical security concepts and hands-on application for students and administrators.
- Future Foundation: Raises awareness and serves as a baseline for developing advanced intrusion detection systems (IDS).

Implementation Results – Core Scripts

- Deauthentication Attack Script (Python 3):
  - Simulates wireless Denial-of-Service (DoS).
  - Injects spoofed 802.11 deauthentication frames to break active client-AP connections.
- Wireless Packet Sniffer (Python 3):
  - Captures and parses live over-the-air traffic.
  - Monitors and flags target authentication and deauthentication signatures.

Lab Environment and Configuration

- Virtual Router Setup:
  - Configured dedicated wireless access point via configuration files.
  - Defined key parameters: SSID, Password, Channel, and WPA2 security.
- Linux Interface Initialization:
  - Crucial Step: Manually provisioned and activated wireless interfaces (e.g., wlan0, wlan1, wlan2) prior to deployment.
  - Result: Virtual router initialized and fully operational.

Security and Authentication Baseline

- WPA2 Encryption:
  - Secures data payloads between the client and the access point.
  - Mitigates unauthorized network eavesdropping.
- Client Authentication (wpa_supplicant):
  - Utilized the open-source wpa_supplicant utility for secure IEEE 802.11 association.
  - Automated connections to the wlan2 virtual access point using client.conf profiles (SSID & Pre-Shared Key).

Live Attack and Forensic Detection

- Real-Time Attack Monitoring:
  - The sniffing module successfully intercepted the deauthentication flood.
  - Provided immediate visibility into anomalous wireless disruptions.
- Wireshark Evidence Validation:
  - Connection Phase: Captured standard client-AP handshake and authentication frames.
  - Attack Phase: Captured matching deauthentication frame signatures injected by the Python script.

User Interface and Telemetry

- Command Line Interface (CLI):
  - Streamlined console output for real-time packet telemetry.
  - Consolidates active monitoring logs, frame headers, and threat alerts into a single view.
  - Baseline State: Displays clean connection logs when no malicious activity is active.

Understanding Wireless Security Concepts

- Wireless Vulnerability: Signals are broadcast openly, allowing attackers to intercept or manipulate unprotected management frames to disrupt users.
- Core Concepts Explored: Practical application of monitor mode, packet sniffing, frame injection, WPA2 authentication, and intrusion detection.
- Technical Insights:
- Kali Linux & Python3: Used for custom attack scripting and GUI development.
- hostapd & Wireshark: Used to simulate access points and analyze abnormal traffic patterns.
- Key Takeaway: Enhanced understanding of wireless vulnerabilities and the necessity of proactive monitoring strategies in cybersecurity.

Project Conclusion

The Wireless Intrusion Detection and Monitoring System (WIDMS) successfully demonstrated Layer-2 security risks in IEEE 802.11 networks by leveraging Kali Linux, Python Scapy, and Wireshark to detect deauthentication floods, probe tracking, and hidden SSID exploits. While currently constrained to a controlled lab environment for threat detection, the project provides a vital educational foundation regarding vulnerable unencrypted management frames. It underscores the power of open-source security monitoring, paving the way for future enhancements such as automated machine learning anomaly detection and active mitigation via Protected Management Frames (802.11w).